

Ethical Issues in Computing

1. **What are ethical issues in computing?**
Ethical issues in computing involve moral challenges such as privacy concerns, data misuse, and ensuring equitable access to technology. They emphasize the responsibility of individuals and organizations to use technology fairly and responsibly.
 2. **Why is privacy an ethical concern in computing?**
Privacy is an ethical concern because personal information can be collected, shared, or misused without consent. This breaches individual autonomy and trust in technology.
 3. **How does digital divide relate to ethical issues?**
The digital divide refers to unequal access to technology, creating ethical concerns about fairness and opportunity. Bridging this gap is essential for ensuring inclusivity.
 4. **What is the role of software developers in addressing ethical concerns?**
Developers should follow ethical guidelines, such as avoiding biased algorithms and protecting user data. They must prioritize transparency and user trust.
 5. **Why is bias in algorithms an ethical issue?**
Biased algorithms can perpetuate discrimination by unfairly favoring or disadvantaging certain groups. Ethical development ensures fairness and inclusivity in technology.
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Intellectual Property Rights (Trademark, Copyright, Patent)

6. **What is a trademark in intellectual property?**
A trademark is a symbol, name, or logo that identifies a brand or product. It helps businesses protect their identity and prevent misuse by others.
 7. **What is the purpose of copyright?**
Copyright protects the original work of creators like books, music, and software. It ensures they control how their work is used and are compensated for their efforts.
 8. **What does a patent protect?**
A patent protects inventions, granting the creator exclusive rights to use, produce, or sell the invention. It encourages innovation by safeguarding creators' interests.
 9. **How do trademarks prevent business conflicts?**
Trademarks establish brand identity, preventing other businesses from using similar marks. This avoids consumer confusion and unfair competition.
 10. **Why is intellectual property protection important in computing?**
IP protection prevents unauthorized use of software, algorithms, and other digital creations. It supports innovation and fair competition in the tech industry.
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Data Protection on the Internet

11. **What is data protection on the internet?**
Data protection ensures personal information is secure from unauthorized access or misuse. It is crucial for maintaining user privacy and trust online.

12. How can encryption help in data protection?

Encryption converts data into unreadable formats, making it accessible only to authorized parties. It safeguards sensitive information from hackers.

13. What are cookies, and how do they relate to data protection?

Cookies store user information on websites, raising data protection concerns. Users must be informed and have control over how cookies track their data.

14. What is the role of GDPR in data protection?

The General Data Protection Regulation (GDPR) enforces strict data privacy rules, giving individuals control over their data and penalizing non-compliant organizations.

15. How can individuals protect their data on the internet?

Using strong passwords, enabling two-factor authentication, and avoiding suspicious links can protect personal data online.

DPA 1998

16. What is the Data Protection Act (DPA) 1998?

The DPA 1998 was a UK law ensuring the lawful processing, storage, and sharing of personal data. It aimed to protect individual privacy.

17. What rights did the DPA 1998 give to individuals?

The Act allowed individuals to access their data held by organizations and request corrections to inaccuracies.

18. How does the DPA 1998 protect personal data?

It required organizations to handle data fairly, securely, and transparently, ensuring proper use and protection against misuse.

19. What replaced the DPA 1998?

The DPA 2018 and GDPR replaced the DPA 1998, providing updated and stronger data protection regulations.

20. Why was the DPA 1998 significant?

It set the foundation for modern data protection laws, addressing the growing concerns about personal data misuse in the digital age.

Spam and Ways to Stop Spam

21. What is spam?

Spam refers to unsolicited messages, often emails, sent in bulk. It disrupts communication and may carry malicious content.

22. Why is spam a problem?

Spam wastes time, consumes resources, and can spread malware or phishing attacks. It undermines productivity and security.

23. What are some ways to stop spam?

Using spam filters, blocking suspicious senders, and avoiding sharing email addresses publicly can help reduce spam.

24. How do spam filters work?

Spam filters analyze email content, sender details, and patterns to detect and block unwanted messages. They help maintain clean inboxes.

25. Why should individuals report spam?

Reporting spam helps email providers improve filters and prevent future spam from similar sources.

Software Piracy and Ways to Avoid Piracy

26. What is software piracy?

Software piracy is the unauthorized copying, distribution, or use of software. It violates intellectual property laws.

27. Why is software piracy a problem?

Piracy leads to revenue losses for developers and may expose users to malware from untrusted sources.

28. How can piracy be avoided?

Using licensed software, promoting awareness about piracy's consequences, and implementing DRM (Digital Rights Management) can prevent piracy.

29. What is the role of DRM in preventing piracy?

DRM restricts how software or digital content can be accessed, copied, or shared, protecting creators from piracy.

30. How does education help reduce piracy?

Educating users about ethical computing and the risks of pirated software encourages responsible behavior and reduces piracy rates.

Cyber Crime

31. What is cyber crime?

Cyber crime involves illegal activities conducted online, such as hacking, fraud, or identity theft.

32. What are some common examples of cyber crime?

Common examples include phishing attacks, ransomware, data breaches, and online scams.

33. Why do cyber crimes occur?

Cyber crimes occur due to financial motives, political agendas, or malicious intent. Weak security systems also enable such crimes.

34. Who are cyber criminals?

Cyber criminals can be organized hacking groups, professional hackers, or disgruntled employees exploiting vulnerabilities.

35. How can individuals prevent cyber crimes?

By using strong security practices like firewalls, regular software updates, and avoiding suspicious links, individuals can minimize the risk of cyber crime.

Hacking

36. What is hacking?

Hacking is the unauthorized access to or manipulation of computer systems. It can be done for ethical or malicious purposes.

37. What are the reasons for hacking?

Reasons include financial gain, political motives, curiosity, or the intent to cause harm.

38. What is the difference between ethical hacking and malicious hacking?

Ethical hacking is done with permission to improve security, while malicious hacking aims to exploit or damage systems.

39. How can organizations prevent hacking?

Organizations can implement strong security protocols, conduct regular audits, and train employees to recognize threats.

40. What is the role of firewalls in preventing hacking?

Firewalls monitor and control network traffic, blocking unauthorized access to systems.

Malicious Intent and Malicious Code

41. What is malicious intent in computing?

Malicious intent refers to the deliberate creation or use of technology to harm others, steal data, or disrupt systems.

42. What is malicious code?

Malicious code is software designed to harm or exploit systems, such as viruses, worms, and ransomware.

43. How does malware spread?

Malware spreads through infected files, emails, or websites, exploiting user trust and system vulnerabilities.

44. Why do attackers create malicious code?

Attackers create malicious code to steal data, disrupt services, demand ransoms, or achieve personal or political goals.

45. How can malicious code be prevented?

Using updated antivirus software, avoiding suspicious downloads, and enabling firewalls can prevent malicious code.

Cyber Criminals

46. Who are organized hackers?

Organized hackers are groups that work together to execute sophisticated cyber attacks, often for financial or political motives.

47. What are professional hackers?

Professional hackers are skilled individuals who exploit systems for monetary gain, often working on the dark web.

48. What role do discontent employees play in cyber crime?

Discontent employees may leak sensitive data or sabotage systems as an act of retaliation against their employers.

49. Why do professional hackers target organizations?

Professional hackers target organizations for financial gains, intellectual property theft, or espionage.

50. How can companies protect themselves from insider threats?

Companies can implement strict access controls, monitor employee activity, and promote a positive work environment.

Prevention of Cyber Laws

51. What is the purpose of cyber laws?

Cyber laws regulate online activities, ensuring digital security and protecting users from cyber crimes.

52. How do cyber laws help in combating cyber crime?

Cyber laws establish legal consequences for cyber crimes, deterring criminals and providing victims with a path to justice.

53. Why is international cooperation important in cyber laws?

Cyber crimes often cross borders, making international cooperation essential for enforcing laws and tracking offenders.

54. What challenges do cyber laws face?

Cyber laws face challenges like jurisdictional conflicts, lack of awareness, and rapidly evolving technologies.

55. How can individuals comply with cyber laws?

Individuals can comply by respecting intellectual property, using secure practices, and avoiding illegal online activities.

Miscellaneous

56. What motivates cyber criminals?

Cyber criminals are often motivated by financial gain, revenge, or ideological beliefs.

57. What is phishing, and why is it dangerous?

Phishing involves tricking individuals into revealing sensitive information. It leads to identity theft and financial losses.

58. What is ransomware?

Ransomware encrypts a victim's data and demands payment for decryption. It disrupts businesses and endangers data.

59. How does social engineering relate to cyber crime?

Social engineering manipulates individuals into divulging confidential information, exploiting human trust instead of technical vulnerabilities.

60. Why is cyber security training important?

Training raises awareness about threats, teaching individuals how to recognize and prevent cyber attacks, reducing risks.